Experiment 10: Towers of Hanoi

Aim:

To solve the Towers of Hanoi problem by generating the optimal sequence of moves to transfer stacked disks following movement constraints.

Algorithm:

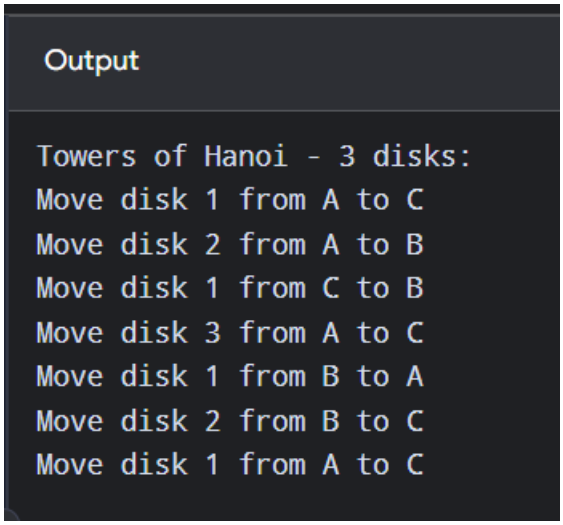
1. To move n disks from source to destination using auxiliary:
   a. Move n-1 disks from source to auxiliary.
   b. Move the nth disk from source to destination.
   c. Move n-1 disks from auxiliary to destination.
2. Base case: n = 1, directly move the disk.

Code:

```
def hanoi(n, source, auxiliary, destination):
    if n == 1:
        print(f"Move disk 1 from {source} to {destination}")
        return
    hanoi(n-1, source, destination, auxiliary)
    print(f"Move disk {n} from {source} to {destination}")
    hanoi(n-1, auxiliary, source, destination)

print("Towers of Hanoi - 3 disks:")
hanoi(3, 'A', 'B', 'C')
```

Sample Output:



Result:

The recursive algorithm correctly generated the optimal 7 moves required to transfer 3 disks from peg A to peg C.